

An IoT-based Smart Adaptive Highway Management System for Noise Pollution Reduction and Traffic Control

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Abstract— Noise pollution is a significant environmental issue in urban areas, predominantly caused by vehicular traffic. Excessive honking significantly contributes to elevated noise levels, adversely affecting residents' health and quality of life. To address this challenge, this study proposes and develops a smart adaptive system to mitigate noise pollution and manage traffic effectively using Internet of Things (IoT) technology. The proposed system integrates IoT devices, such as LM393 sound sensors, NodeMCU ESP8266, and communication modules, to monitor noise levels at key intersections. Real-time data from these sensors are processed and analyzed to identify instances of excessive honking. When noise levels exceed predefined thresholds, the system initiates corrective measures to mitigate noise pollution and enhance traffic flow. Traffic signals are dynamically controlled based on detected noise levels, with penalties imposed for repeated instances of excessive honking during red signal phases. These penalties include extending the red signal duration as a deterrent to discourage further honking. Additionally, the system provides visual feedback to drivers via LED displays and LCD screens, indicating the current traffic signal state and countdown timers.

Keywords—IoT, Adaptive traffic control, Noise pollution reduction, Microcontroller.

I. INTRODUCTION

In contemporary urban environments, the increasing proliferation of vehicular traffic has led to numerous challenges, including noise pollution and traffic congestion, both of which significantly affect the quality of life and well-being of residents [1]. Among these, excessive honking has become a major contributor to elevated noise levels, with detrimental effects on public health and overall urban livability [2–3]. Furthermore, traffic congestion not only causes delays and frustration among commuters but also leads to increased fuel consumption and environmental degradation [4]. According to the World Health Organization (WHO), noise pollution is one of the most harmful environmental factors affecting human health [5–6]. It is estimated to cause over 72,000 hospital admissions and nearly 16,600 deaths

annually [7]. Prolonged exposure to high levels of noise can lead to a range of health issues, including hearing loss, cardiovascular problems, sleep disturbances, and increased stress levels [8]. Noise pollution not only affects humans but also disrupts wildlife, disturbing their natural habitats and communication patterns. Additionally, traffic congestion is a pervasive issue in many urban areas, leading to delays, frustration among commuters, increased fuel consumption, and heightened air pollution [9].

To address these pressing issues, there is a growing need for innovative solutions that leverage emerging technologies to develop smarter and more sustainable transportation systems. In this context, an IoT-based smart adaptive highway management system for noise pollution reduction and traffic control offers a proactive approach to mitigating noise pollution and improving traffic management along highways. Traffic mismanagement can lead to accidents, and excessive horn noise poses significant health risks. In [10], the authors proposed a modular, scalable real-time noise monitoring system using an ESP8266 microcontroller and a low-cost calibrated sound sensor, aimed at improving health and well-being via a web platform and mobile application. Other IoT-based systems include methods for monitoring air quality and noise using low-cost sensors with a Raspberry Pi and cloud storage via GPRS. Additionally, a platform utilizing Narrow Band IoT technology incorporated MEMS microphones calibrated through three different methods. A noise level detector with a 16x2 LCD screen, LED indicators, and audio notifications was developed to monitor noise levels [11]. The device triggers a red LED for noise above 75 dB, a yellow LED with a beep for levels above 56 dB, and indicates calm conditions below 55 dB. The system effectively detects and displays noise levels between 50 and 100 dB, providing audio-visual alerts for monitoring and measurement.

In our work, we make the following contributions:

- Developed a smart adaptive system integrating LM393 sound sensors, NodeMCU ESP8266, and communication modules for real-time monitoring and analysis of noise levels.
- Implemented traffic signal adjustments based on noise levels, with penalties such as extended red signal durations to deter excessive honking.
- Provided audio-visual feedback to drivers using LED displays and LCD screens, and displaying signal states.
- Aimed to reduce noise pollution and improve urban residents' health and quality of life.

The rest of the manuscript is organized as follows: Section II discusses the three key components of the project: NodeMCU ESP8266, LM393 Sound Sensor, and LCD Display. Section III presents the conceptual block diagram, circuit diagram, and a detailed explanation of the operating principles, including the final real-life prototype and its outcome analysis. Finally, Section IV provides the concluding remarks.

II. KEY COMPONENTS

The primary key components of the developed prototype include the NodeMCU ESP8266, LM393 Sound Sensor, and LCD Display. The NodeMCU ESP8266 serves as the central microcontroller, enabling Wi-Fi connectivity for seamless data transmission and communication with other devices. The LM393 Sound Sensor is responsible for detecting noise levels, providing real-time input on sound intensity to the system. The LCD Display serves as the visual interface, presenting relevant information such as current noise levels, traffic signal status, and countdown timers, allowing users to monitor and interact with the system effectively.

A. Node MCU ESP8266

NodeMCU is an open-source IoT platform based on the ESP8266 microcontroller, specifically the ESP-12E version. It facilitates data communication between sensors and Firebase. The NodeMCU features both analog and digital I/O pins, enabling seamless integration with sensors such as the HCSR04 and MQ-2. It boasts an extended number of GPIO pins and includes an onboard Digital-to-Analog Converter (DAC), allowing it to read analog sensor values. Unlike the ESP-01 version, NodeMCU operates at a higher processing frequency and supports the integration of multiple sensors. Fig. 1 illustrates the pin configuration of the NodeMCU ESP8266.

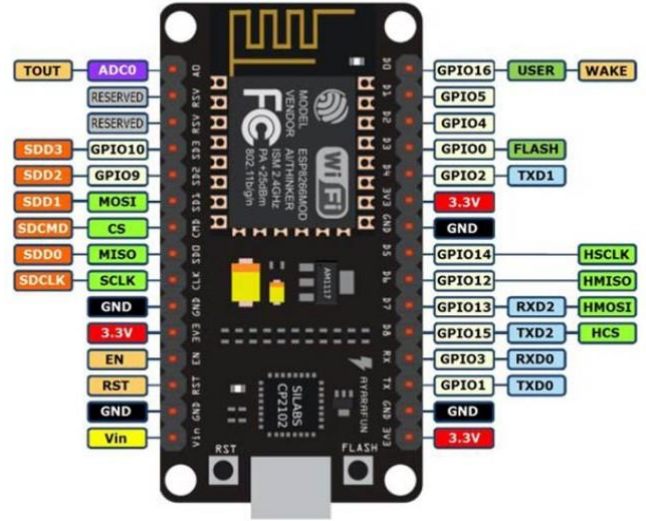


Fig. 1: Node MCU ESP8266

B. LM393 Sound Sensor

The LM393 is a dual voltage comparator IC, which means it has two separate comparators inside. Each comparator has two inputs (inverting and non-inverting) and one output. Here's a basic overview of the pin functions of the LM393:

Vcc (+): This is the positive supply voltage pin. It typically ranges from 2 to 36 volts, depending on the specific model of the LM393.

Ground (-): This pin is connected to the ground (0 volts) of the circuit.

Inverting Input (IN-): This pin is the inverting input of the first comparator. It's connected to the signal source you want to compare.

Non-Inverting Input (IN+): This pin is the non-inverting input of the first comparator. It's typically connected to a reference voltage against which the input signal is compared.

Output (OUT1): This pin is the output of the first comparator. The output voltage changes based on the comparison between the voltages applied to the inverting and non-inverting inputs.

Inverting Input (IN-): This pin is the inverting input of the second comparator. It's connected to the signal source you want to compare.

Non-Inverting Input (IN+): This pin is the non-inverting input of the second comparator. It's typically connected to a reference voltage against which the input signal is compared.

Output (OUT2): This pin is the output of the second comparator. Like OUT1, the output voltage changes based on the comparison between the voltages applied to the inverting and non-inverting inputs.

To create a sound sensor using the LM393, we would typically connect the output of a sound sensor (such as a microphone) to one of the comparator inputs, and a reference

voltage (often set using a voltage divider with resistors) to the other input. Depending on the sound level detected, the output of the comparator will change, which can be used to trigger other components or circuits. Remember to consult the datasheet of the LM393 for detailed information on its pinout, recommended operating conditions, and application circuit examples. Fig. 2 shows the pin configuration of a LM393 Sound Sensor.

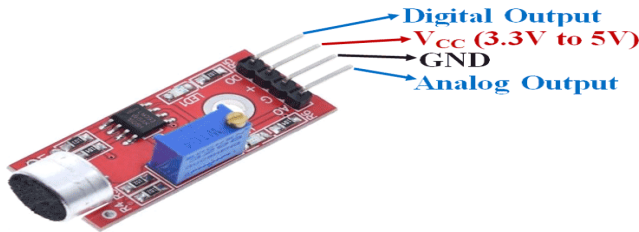


Fig. 2: LM393 Sound Sensor

C. 16×2 LCD Display

An LCD (Liquid Crystal Display) module typically consists of several pins, each serving a specific function to interface with a microcontroller or other electronic devices. Here's a common set of pins found on most character LCD displays (such as 16x2 or 20x4):

VSS (Ground): This pin connects to the ground (0V) of the circuit.

VDD or VCC (Power Supply): This pin is connected to the positive supply voltage, usually +5V in most applications.

VO (Contrast Adjustment): This pin is used to adjust the contrast of the display. Applying a voltage to this pin adjusts the contrast of the characters displayed.

RS (Register Select): This pin is used to select between data register (when logic high) and instruction register (when logic low).

RW (Read/Write): This pin is used for selecting the read or write mode. It is typically grounded for write operations, meaning data is being sent to the LCD.

E (Enable): This pin is used to enable the data transfer. Rising edge of the pulse is used to latch the data present on the data lines.

D0-D7 (Data Lines): These pins (sometimes referred to as DB0-DB7) are used to send data or commands to the LCD. In 4-bit mode, typically only D4-D7 pins are used.

A (Anode): This pin is for the backlight's anode connection (positive terminal). It is connected to a positive voltage for backlighting.

K (Kathode): This pin is for the backlight's cathode connection (negative terminal). It is usually connected to the ground for backlighting.

Depending on the specific LCD module or manufacturer, some pins may have additional functions or different names. Additionally, there may be extra pins for features like backlight control, power supply for the backlight, etc. Fig. 3 shows the pin configuration of a 16*2 LCD Display.

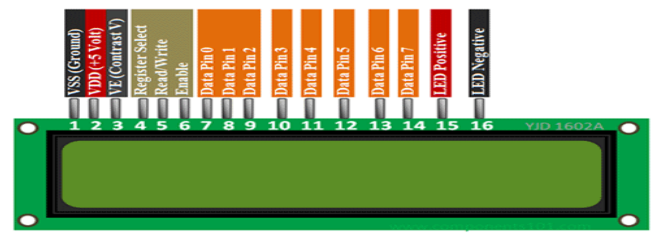


Fig. 3: 16*2 LCD Display

III. CONCEPTUAL PRESENTATION, CIRCUIT DIAGRAM AND WORKING MECHANISM OF THE PROPOSED MANAGEMENT SYSTEM

In urban areas, noise pollution and traffic congestion are significant concerns that affect the quality of life for residents. To address these issues, an integrated system leveraging Internet of Things (IoT) technology can provide real-time monitoring of noise levels and traffic flow, enabling dynamic management for a more efficient and environmentally friendly urban environment. Fig. 4 shows the conceptual diagram of the proposed IoT-based smart adaptive highway management system, which is designed to reduce noise pollution and improve traffic control. The system uses Noise Pollution Detection Sensors placed in key locations to monitor noise levels in real time. These sensors send their data to a Data Processing Unit, which organizes and prepares the information for analysis. A Microcontroller or Microprocessor then evaluates the noise data and initiates necessary actions if the levels exceed predefined limits. Traffic Sensors are also integrated into the system to monitor vehicle flow, speed, and congestion, providing critical insights into traffic conditions. The data collected is formatted by a Data Transmission module to make it suitable for internet-based communication. To improve the system further, a Feedback Mechanism allows users to report incidents of noise pollution and offer suggestions. A Communication Network, whether wired or wireless, connects all the system components, such as sensors, actuators, gateways, and cloud servers, ensuring smooth data exchange. Finally, Internet Connectivity enables the transfer of this data between local networks and cloud platforms, allowing for remote monitoring and adaptive management of highways. Fig. 5 presents the schematic circuit diagram of the IoT-based smart adaptive highway management system. The process can be summarized as follows:

1. Noise sensors continuously measure ambient noise levels.
2. The microcontroller processes the data and communicates with the IoT module.
3. The IoT module transmits the data to the cloud-based traffic management system.
4. The traffic management system analyzes the data and adjusts traffic signal timings.
5. Traffic Signal Controllers implement the adjustments.
6. LED displays at intersections show the current traffic signal state.

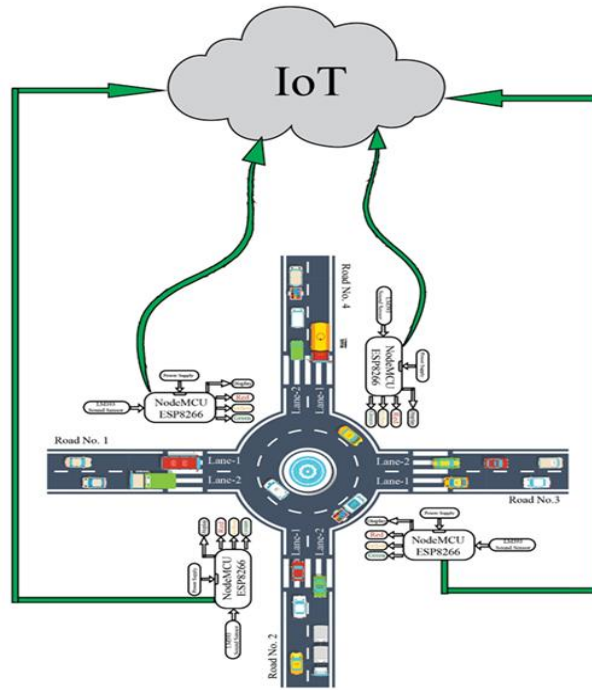


Fig.4: Conceptual Diagram of Proposed IoT-based Smart Adaptive Highway Management System

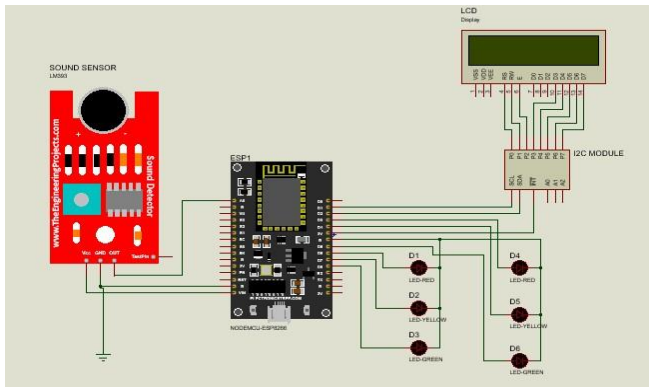


Fig-5: Schematic Circuit Diagram of the Proposed IoT-based Smart Adaptive Highway Management System

The schematic provides a foundational structure for implementing an IoT-based smart adaptive highway management system aimed at reducing noise pollution and controlling traffic. Depending on specific requirements, additional components like more advanced sensors, machine learning modules, and enhanced communication protocols may be integrated. Fig. 6 presents a flowchart of the designed IoT-based smart adaptive highway management system for noise pollution reduction and traffic control at a four-road intersection. The flowchart outlines the process, starting with data collection and proceeding through analysis, decision-making, and action implementation.

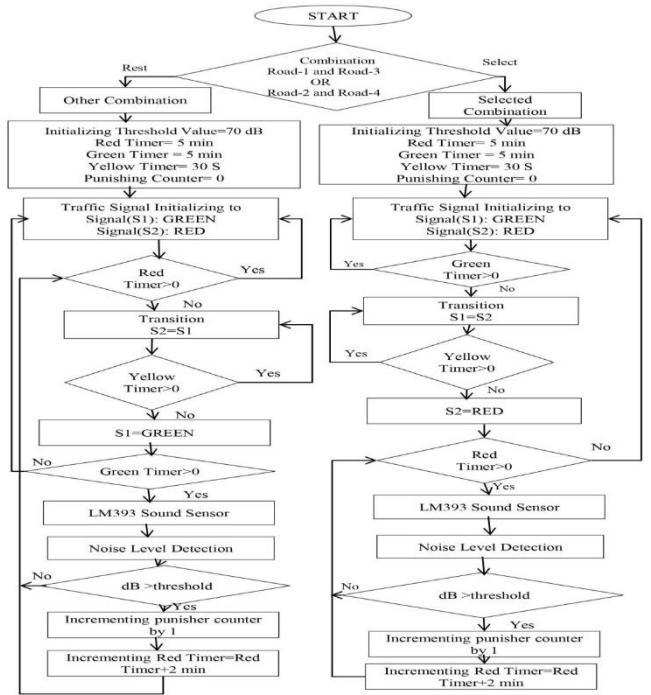


Fig. 6: Flow Chart of the Proposed IoT-based Smart Adaptive Highway Management System at a Four-road Intersection

The system begins by initializing and detecting the status of all roads, identifying which ones are active or inactive. Noise and traffic sensors on the active roads collect data on noise levels, while disregarding data from inactive roads. The collected data from the active roads is transmitted to a central processing unit (CPU) via IoT communication protocols.

The CPU processes this data to evaluate noise pollution levels and traffic conditions. The system analyzes the

processed data to determine if noise pollution exceeds predefined thresholds and if traffic congestion is present on the active roads, ignoring data from inactive roads in the analysis. Based on this analysis, the system decides on appropriate control actions. Actuators adjust traffic lights and road signs on the active roads according to the control actions. Variable message signs (VMS) and other communication channels inform drivers on the active roads about the changes and recommended actions. The system incorporates a feedback loop, continuously monitoring the impact of the implemented measures on the active roads and making real-time adjustments based on feedback from sensors.

The system monitors noise levels using sensors. When the noise levels exceed a specified threshold (for example 65 dB), indicating excessive honking, the system initiates a penalty. If the noise levels exceed the threshold while the traffic signal is in the red state, a penalty is imposed by adding 2 minutes to the red signal countdown timer. This serves as a punishment for unnecessary honking. The traffic signal prototype consists of three LEDs representing green, yellow, and red states. The green signal is active for 5 minutes on the main road and 5 minutes on the side road, followed by a 30-second transition to the red signal. The red signal remains active for 5 minutes on the main road and 5 minutes on the side road. A countdown timer for each state is displayed on an LCD screen. The penalties serve as a deterrent for unnecessary honking. By adding extra time to the red signal, commuters experience delayed progress, discouraging them from honking excessively.

User interaction is facilitated through mobile apps and websites that provide real-time updates about traffic conditions and noise-related advisories on the active roads, along with notifications about the inactive status of the other two roads. Periodic reports are generated, detailing the system's performance with a focus on the active roads while noting the inactivity of the other roads. These reports are reviewed by traffic authorities to assess the effectiveness of the measures and to plan for future improvements, ensuring the system adapts efficiently to the current highway conditions.

IV. PROTOTYPE DEVELOPMENT AND TESTING OF THE PROPOSED MANAGEMENT SYSTEM

To implement real-time noise level monitoring and traffic control for an IoT-based smart adaptive highway management system, as depicted in Fig. 7, follow these steps:

1. Set Up ThingSpeak:

- Create an account on the ThingSpeak platform.
- Create a new channel dedicated to storing noise level data.
- Configure the channel with appropriate fields. For this application, create a field specifically for noise level measurements.

2. Hardware Configuration:

- Connect LM393 sound sensors to NodeMCU ESP8266 microcontrollers. The LM393 sensors are used to detect ambient noise levels and output an analog signal proportional to the sound intensity.

- Wire the LM393 sensor's output to an analog input pin on the NodeMCU ESP8266. Ensure proper power and ground connections are established.

3. Programming the ESP8266:

- Write a program for the ESP8266 to read analog values from the noise sensor.
- Convert the raw analog signal into meaningful noise level measurements.
- Use the ESP8266's Wi-Fi capability to establish an internet connection.
- Transmit the noise level data to the ThingSpeak channel at regular intervals using the ThingSpeak API.

4. Data Visualization and Analysis:

- Configure ThingSpeak to store the incoming noise data in real time.
- Use ThingSpeak's built-in visualization tools to plot and monitor noise levels for analysis.
- Optionally, process the data using MATLAB integration on ThingSpeak for advanced analytics.

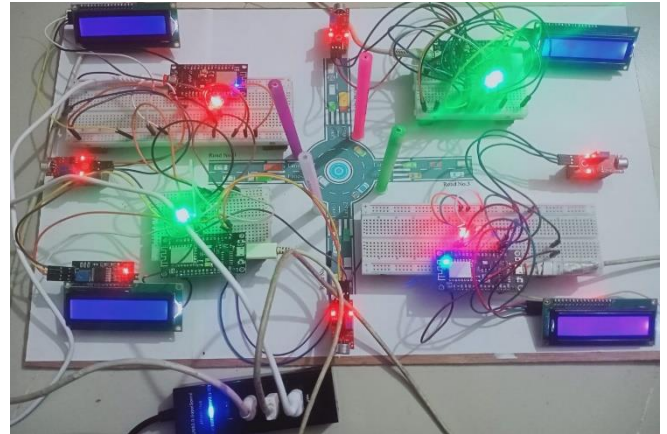


Fig. 7: Project Prototype of the Proposed IoT-based Smart Adaptive Highway Management System

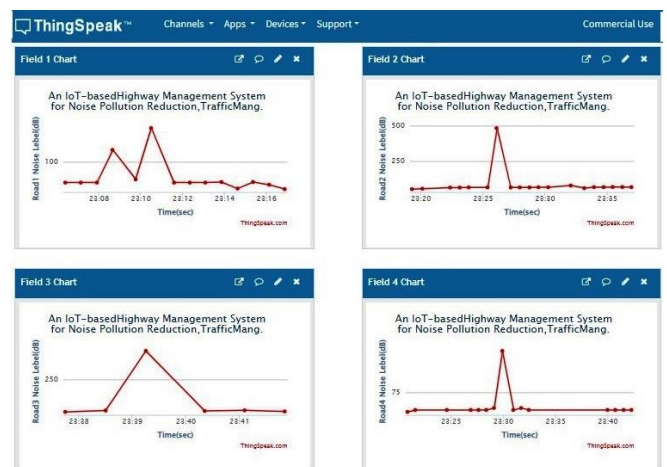


Fig. 8: ThingSpeak Real Time Noise Level

An example of connecting the ESP8266 microcontroller to the LM393 noise sensor and uploading data to ThingSpeak is

illustrated in Fig. 8. The microcontroller periodically reads the sensor data, processes it, and sends it to the ThingSpeak platform for real-time visualization and adaptive highway management decision-making. Fig. 9 illustrates the three possible stages of noise levels on a road:

Case 1: Normal

This state indicates that the noise level on the road is below the predefined threshold, representing typical and acceptable noise conditions.

Transition State

This intermediate state occurs when the noise level is approaching the predefined threshold but has not yet exceeded it. It acts as a buffer state, providing an early indication of increasing noise levels that may escalate further.

Case 2: Honk

This state occurs when the noise level exceeds the predefined threshold, signaling higher-than-normal noise levels, typically caused by excessive honking or other loud disturbances.

These stages allow for a more nuanced monitoring and categorization of road noise levels, enabling adaptive highway management to respond proactively to changes in noise conditions.



Case 1: Normal



Transition State



Case 2: Honk

Fig. 9: Three possible stages of noise levels on a road

V. CONCLUSION

In conclusion, the implementation of an IoT-based system for noise pollution detection and traffic management offers a promising solution to address two critical urban challenges simultaneously. By integrating IoT technology with sensors for noise detection and traffic monitoring, the system enables real-time data collection and analysis, facilitating more efficient urban management. Strategically positioned noise sensors allow continuous monitoring of noise levels across the city, helping identify pollution hotspots and assess the impact of various noise sources, thereby informing targeted interventions. Traffic management sensors provide insights into traffic flow, congestion, and vehicle counts, enabling authorities to optimize traffic flow, implement control measures, and enhance road safety. The IoT connectivity ensures seamless communication between sensors, data processing units, and central control systems, supporting real-time monitoring and responsive actions. Cloud-based data storage and analytics offer long-term trend analysis, predictive modeling, and data-driven decision-making. Ultimately, this integration of IoT technology provides a holistic approach to urban planning, promoting livable, sustainable, and resilient cities while enhancing the quality of life for residents and visitors alike.

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